

Detecting Cross-Turn Rhetorical Fallacies in Discourse: Motte-and-Bailey Claim-Shift and Strawman Mischaracterization

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Abstract

Existing computational fallacy detection treats fallacies as a property of a single text span: a benchmark such as MAFALDA (Helwe et al., 2024) or the fallacy classifiers trained on political debate transcripts (Goffredo et al., 2022, 2023) label one utterance as containing (or not containing) a named fallacy type. A distinct and, to our knowledge, computationally unaddressed class of rhetorical move is *relational*: it exists only in the relationship between two utterances produced at different points in a discourse, and is invisible to single-span classification. We propose to formalize and detect two such relational fallacies – the **motte-and-bailey** claim-shift, in which a speaker retreats from a controversial claim to a weaker, easily-defensible one under challenge while treating the two as equivalent, and **strawman** mischaracterization, in which a respondent argues against a distorted version of a claim the original speaker did not make. We propose a two-stage pipeline – claim extraction followed by a claim-pair relation classifier – trained and evaluated on a purpose-built dataset of political debate transcripts and moderated online discussion threads, combining human annotation with LLM-assisted augmentation in the style of prior fallacy-dataset construction (Jin et al., 2024). We discuss concretely where this data will come from, why constructing it is the central difficulty of the project, and what off-the-shelf tooling we will and will not rely on.

1 Motivation

Computational argument mining has produced substantial infrastructure for detecting logical fallacies in text: MAFALDA (Helwe et al., 2024) unifies over 30 fallacy types into a shared taxonomy; Goffredo et al. (2022) and Goffredo et al. (2023) detect and classify fallacies in political debate transcripts; Jin et al. (2024) scale dataset construction using LLM-assisted crowd annotation; and Ruggeri et al.

(2025) address the inherent subjectivity of fallacy judgments through human label variation.

A structural limitation runs through all of this work: every task above is defined at the level of a single utterance or short text span. The model is asked, “does this piece of text contain a fallacy,” and the fallacy – if present – is a self-contained property of that text. This framing cannot represent an important and common class of rhetorical move in real discourse, where the violation is not in any one statement but in the *relationship* between two statements made at different points by the same or different speakers. The motte-and-bailey maneuver is the clearest example: taken alone, a speaker’s retreat to a modest claim under challenge is entirely reasonable – the fallacy only exists when that retreat is later treated as having defended the original, bolder claim. A single-span classifier evaluating either statement in isolation has no way to detect this, because neither statement is individually fallacious. The same is true of strawman argumentation, where the fallacy is a mismatch between what was claimed and what was rebutted, not a property of the rebuttal text alone.

A second, more practical shortcoming compounds the first: because no benchmark treats claim pairs as the unit of annotation, no dataset exists that even represents the relational task, let alone labels it. This is not a modeling gap that a bigger single-span classifier can close – it is a representational gap in how the data itself is structured. Any project in this space must therefore treat *dataset construction* as a first-class deliverable rather than a preliminary step, and must be explicit about where usable raw discourse can still be obtained now that several previously-standard sources (notably live Reddit scraping, discussed in Section 5) have become substantially harder to access than when earlier corpora such as CMV (Tan et al., 2016) were built.

This is the shortcoming our project addresses: existing fallacy detection is single-span by con-

struction, existing discourse corpora are not organized around claim pairs, and a genuinely open, formally under-specified task – relational fallacy detection across discourse turns – falls outside what current systems can represent or existing benchmarks can evaluate.

2 Related Work and Background

Fallacy detection. MAFALDA (Helwe et al., 2024) and the political-debate fallacy classifiers of Goffredo et al. (2022, 2023) establish the dominant taxonomy-and-span-classification paradigm we depart from. Jin et al. (2024) demonstrate that LLM-assisted crowdsourcing can produce fallacy-labeled data at scale for news comments, which we adopt as a methodological precedent for our own dataset construction given the absence of any existing labeled data for the relational task we propose. Ruggeri et al. (2025) show that fallacy judgments carry genuine annotator disagreement rather than a single ground truth, which directly motivates our annotation protocol (Section 5).

Argumentation theory. We ground our claim representation in two established formalisms rather than inventing an ad hoc one. Toulmin’s model of argument (Toulmin, 1958) decomposes an argument into claim, evidence, and warrant, giving us a principled unit (“claim”) to extract and compare across turns. Dung’s abstract argumentation frameworks (Dung, 1995) represent arguments as nodes in a graph with attack/support relations between them; we treat a motte-and-bailey shift or a strawman mischaracterization as a specific, detectable *edge type* between two claim-nodes, rather than a property of a node. This graph view is what motivates casting our task as *claim-pair* classification rather than an extension of span classification.

Natural language inference and off-the-shelf models. We use NLI-style entailment scoring (Bowman et al., 2015) as the core signal for our relation classifier: a motte-and-bailey shift is characterized by claim B being a strictly weaker, narrower statement that does not entail the original claim A, while being offered as though it does; a strawman is characterized by low entailment between the claim actually made and the claim being rebutted. Rather than training an entailment model from scratch, which is infeasible given our data budget, we plan to reuse a publicly released, multi-corpus-trained NLI model in the family described

by Laurer et al. (2024) (trained jointly on MNLI, FEVER-NLI, ANLI, LingNLI, and WANLI) as a frozen entailment/contradiction scorer, treating entailment-model reuse the same way prior fallacy work reuses pretrained LLMs rather than training taxonomy-specific encoders from zero.

Conversational and discourse toolkits. ConvoKit (Chang et al., 2020) packages several conversational corpora, including the Winning Arguments / CMV corpus of Tan et al. (2016), with a uniform Python API for traversing discourse trees, reply chains, and per-utterance metadata. We use this as our primary discourse-traversal layer for the online-discussion side of our data (Section 5) rather than writing custom Reddit-tree parsing code.

3 Contribution

We make three contributions. First, we formalize **relational fallacy detection** as a task distinct from single-span fallacy classification: given an ordered pair of claims linked by discourse context (a challenge, in the motte-and-bailey case; a rebuttal, in the strawman case), predict whether the pair exhibits the relevant relational fallacy. This directly addresses the representational shortcoming in Section 1: by making the claim pair, rather than the utterance, the unit of prediction, the task becomes capable of expressing violations that single-span labels structurally cannot. Second, we propose a **claim-pair relation classifier** that combines NLI-based entailment/specificity scoring between claim pairs with discourse-position features (e.g., whether claim B follows a challenge to claim A, speaker identity across the pair), which is – to our knowledge – the first model architecture aimed specifically at this relational class of fallacy rather than single-span classification. Third, and most directly responding to the data-availability gap identified in Section 1, we construct and release a **seed dataset** of hand-annotated claim pairs drawn from political debate transcripts and moderated discussion threads, augmented with LLM-generated synthetic examples following Jin et al. (2024), since no existing dataset labels this task and the sources we use are chosen specifically because they remain accessible under current data-access constraints (Section 5).

4 Proposed Methodology

Our pipeline has two stages, described below, followed by the concrete tooling choices behind each.

Stage 1 – Claim extraction. For each discourse turn, we extract the central claim(s) using a prompted LLM extractor, following the claim/evidence decomposition of Toulmin’s model (Toulmin, 1958). The extractor is prompted to output a normalized, decontextualized claim string per turn (resolving pronouns and ellipsis against the immediately preceding turn), since downstream entailment scoring is sensitive to dangling references. We evaluate extraction quality against a small hand-annotated subset (exact-match and semantic-similarity agreement against gold claim strings) before proceeding to Stage 2, since downstream relation classification is only as reliable as the claims it operates over.

Stage 2 – Claim-pair relation classification. For each candidate pair of claims (A, B) linked by discourse context, we compute three feature groups: (i) bidirectional NLI entailment/contradiction/neutral scores between A and B from the frozen DeBERTa-v3-based NLI model described in Section 2; (ii) a specificity/hedging delta capturing whether B is a narrower or more qualified version of A , computed from surface hedging cues (modal verbs, quantifier scope, explicit qualifiers) together with the NLI asymmetry between $A \Rightarrow B$ and $B \Rightarrow A$; (iii) discourse-position features – turn distance between A and B , whether an intervening turn explicitly challenged A , and whether A and B share a speaker (motte-and-bailey is typically same-speaker; strawman is typically cross-speaker). These features feed a classifier predicting one of *motte-and-bailey shift*, *strawman mischaracterization*, or *no relational fallacy*; we will compare a lightweight gradient-boosted tree model (LightGBM/XGBoost over the engineered features) against a fine-tuned transformer that consumes A , B , and the discourse features jointly, to measure how much the engineered discourse-position features contribute beyond what a transformer can infer from text alone.

5 Dataset and Data Acquisition

No existing dataset labels this task, so data construction is itself part of the project and, in our judgment, its central difficulty – the “moat” of this proposal is not the classifier architecture but assembling discourse where challenge-and-retreat or claim-and-rebuttal pairs are frequent, identifiable, and legally/practically obtainable at the scale of a course project. We plan two primary sources,

chosen specifically because both remain accessible without live platform scraping.

Political debate transcripts. Following the precedent of Goffredo et al. (2022, 2023), we will draw transcripts from the American Presidency Project at UC Santa Barbara (University of California, Santa Barbara), a public-domain archive of U.S. presidential, primary, and vice-presidential debate transcripts spanning 1960–present. Debate transcripts are attractive for this task because moderator- and opponent-driven challenges to a claim, followed immediately by a narrower restatement, occur densely and are easy for annotators to locate by searching around moderator follow-up questions.

Moderated online discussion threads. For informal, multi-party discourse we use the Winning Arguments / r/ChangeMyView corpus (Tan et al., 2016), accessed through ConvoKit (Chang et al., 2020) rather than the live Reddit API. This choice is deliberate: Reddit’s 2023 API pricing changes closed off free, high-volume programmatic access to live subreddit data, and Pushshift – the community mirror many prior NLP projects relied on for historical Reddit data – no longer offers the open, unauthenticated bulk access it once did. Building a new live scraper for this project would therefore add legal and rate-limit risk with no benefit, since a large, already-cleaned, conversation-threaded CMV snapshot is available as a static release through ConvoKit and is exactly the genre (sustained claim defense under repeated challenge, explicitly framed around changing one’s stated position) that motte-and-bailey shifts occur in. We considered structured debate platforms such as Kialo as a further source, but rejected it for the seed set: Kialo’s argument trees are pre-structured by contributors into discrete pro/con nodes, which removes much of the naturalistic hedging and claim-drift language our task depends on; we may revisit a small Kialo-derived supplementary set only if seed-set volume proves insufficient.

Annotation. We will hand-annotate an initial seed set (target: 150–250 claim pairs) using POTATO (Pei and Jurgens, 2023), a lightweight, configurable web-based text annotation tool, so annotators see the full surrounding discourse context (not just the isolated claim pair) when labeling. Work is split across the team of four annotators against a shared guideline document that gives

worked positive and negative examples of each fallacy type, since the two phenomena are easy to conflate with ordinary claim refinement or legitimate paraphrase. We will then augment the seed set with LLM-generated synthetic claim pairs in the style of Jin et al. (2024), using the seed set’s positive examples as few-shot templates and manually spot-checking a sample of generations for label validity before adding them to training data. Given the annotator disagreement documented for even single-span fallacy judgments (Ruggeri et al., 2025), we will report inter-annotator agreement (Cohen’s/Krippendorff’s α over a doubly-annotated subset) and adopt a label-variation-aware evaluation scheme rather than collapsing to a single ground truth.

6 Evaluation Plan

We compare against two baselines: (i) zero-shot LLM prompting on the concatenated pair with no structural features, to establish whether relational structure adds value over raw language understanding alone; (ii) a naive application of an existing single-span fallacy classifier (applying MAFALDA-style span labels to each claim independently and flagging a pair only if either span is independently flagged) to demonstrate empirically that single-span methods cannot detect relational fallacies by construction, since by definition neither claim in a true motte-and-bailey or strawman pair need be independently fallacious. We additionally report the standalone accuracy of the frozen NLI entailment score as a third, feature-ablation baseline, to isolate how much the discourse-position and specificity features (Section 4) add over entailment alone. We report per-class F1 and macro-F1 on held-out annotated pairs, and where annotator disagreement is high, we report agreement-weighted metrics following Ruggeri et al. (2025). We will additionally report a confusion analysis distinguishing true relational fallacies from benign claim refinement (legitimate narrowing without equivocation) and benign disagreement (accurate rebuttal of the actual claim), since these are the most likely sources of annotator and model error.

7 Tentative Timeline

Weeks 1–3: claim extraction pipeline and validation; annotation guideline finalized; ConvoKit/CMV and UCSB debate transcript pulls completed. Weeks 4–6: seed dataset annotation via

POTATO and LLM-augmented expansion. Weeks 7–9: relation classifier implementation and baseline comparisons, aligned with the Mid-Submission deadline (30 September). Weeks 10–12: error analysis, ablations, and write-up toward the Final Submission (31 October).

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